Monge and Kantorovich Optimal Transport

From matchings to probability measures, maps, relaxations, and Wasserstein distances.

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Optimal Transport for Machine Learners

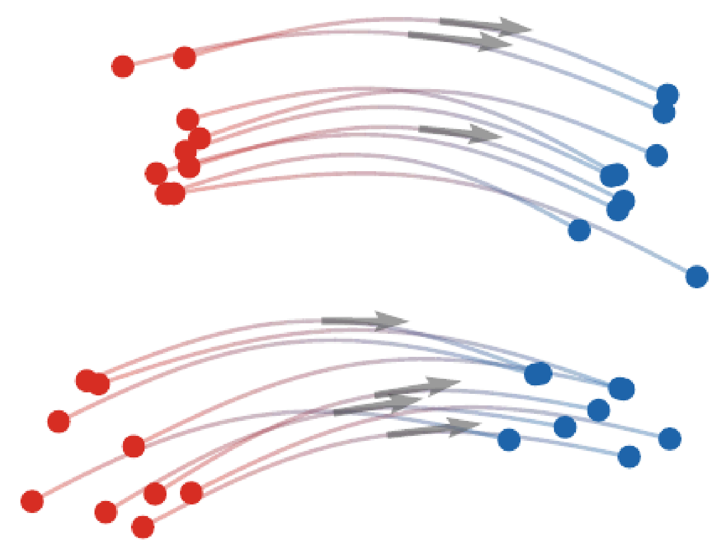
Why compare distributions?

Modern machine learning repeatedly asks for distances between probability laws:

- empirical data versus a model distribution,
- source versus target domains,
- generated samples versus real samples,
- evolving particle clouds during training.

Optimal transport compares distributions by using the geometry of the ambient space.

source: α, x, a **target:** β, y, b **transport:** T, π, P



Transport turns distribution comparison into geometry.

The fitting problem

Data and model distributions.

A model produces samples from α_θ, while observations come from a target law β. A geometric fitting objective has the form

$$\min_\theta D(\alpha_\theta, \beta),$$

where D is a discrepancy between probability measures.

In practice, both laws are often empirical:

$$\alpha_\theta^n = \frac{1}{n}\sum_{i=1}^{n}\delta_{x_i(\theta)}, \qquad \widehat{\beta}_m = \frac{1}{m}\sum_{j=1}^{m}\delta_{y_j}.$$

If $x_i(\theta) = g_\theta(z_i)$ for latent samples $z_i \sim \zeta$, this is the push-forward model

$$\alpha_\theta = (g_\theta)_\#\zeta, \qquad \min_\theta D((g_\theta)_\#\zeta, \beta).$$

OT chooses D by asking how much effort is needed to move the mass of α_θ onto β.

Computational OT in practice

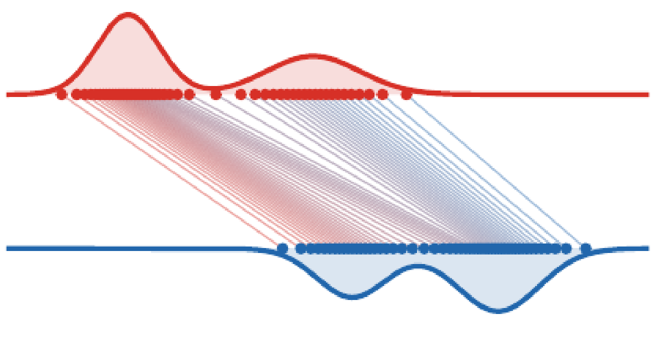
OT is not a single formula, but a computational toolbox.

Computational ecosystem.
The same geometric principle gives sorting algorithms in one dimension, assignment algorithms for equal weights, linear programs for general discrete measures, semi-discrete solvers for continuous-to-discrete maps, and Sinkhorn iterations for differentiable large-scale losses.

This course uses the POT library and the numerical figures of the OT4ML book to keep these regimes visible throughout.

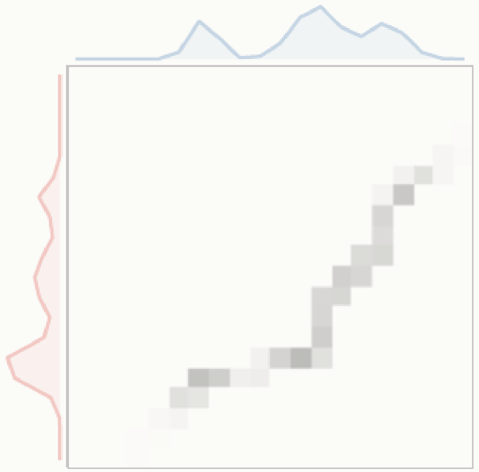
Roadmap: discrete Monge matching

1. Monge matching



2. Continuous Monge

3. Kantorovich relaxation



4. Metric structure

5. Applications

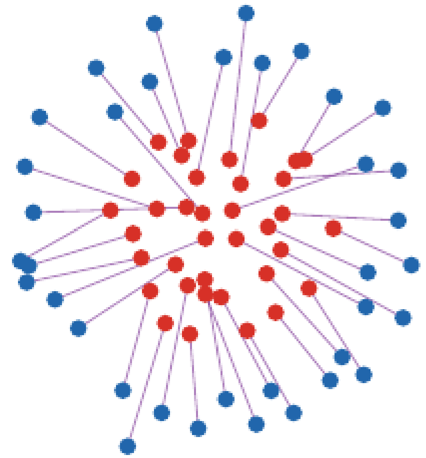
Finite Monge problem

Let $x_1, \dots, x_n$ and $y_1, \dots, y_n$ be two point clouds and let $c_{i,j} = c(x_i, y_j)$.

Permutation matching.
The finite Monge problem is

$$\min_{\sigma \in \mathfrak{S}_n} \sum_{i=1}^{n} c_{i,\sigma(i)}.$$

For $c(x,y) = \|x - y\|^p$, it matches points while paying a geometric displacement cost.



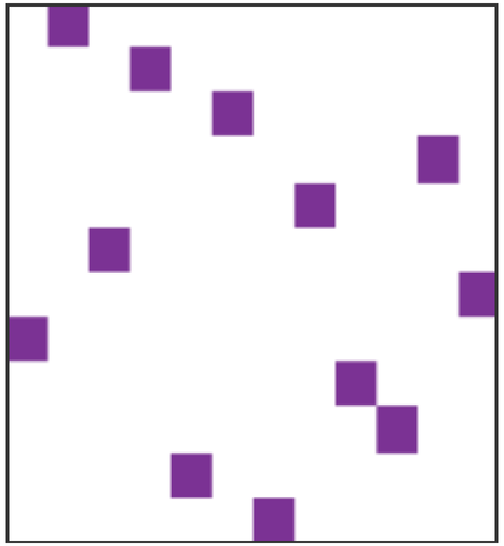
Equivalently, one minimizes $\langle C, P_\sigma \rangle$ over permutation matrices $(P_\sigma)_{i,j} = \mathbf{1}_{\{j=\sigma(i)\}}$.

Why this is hard

Combinatorial nature.
The feasible set has $n!$ permutations. A brute force search is impossible except for very small n.

For arbitrary cost matrices, this is the linear assignment problem. Classical methods include:

- Hungarian algorithm,
- auction algorithm,
- linear programming and network simplex variants.



A matching is a permutation matrix.

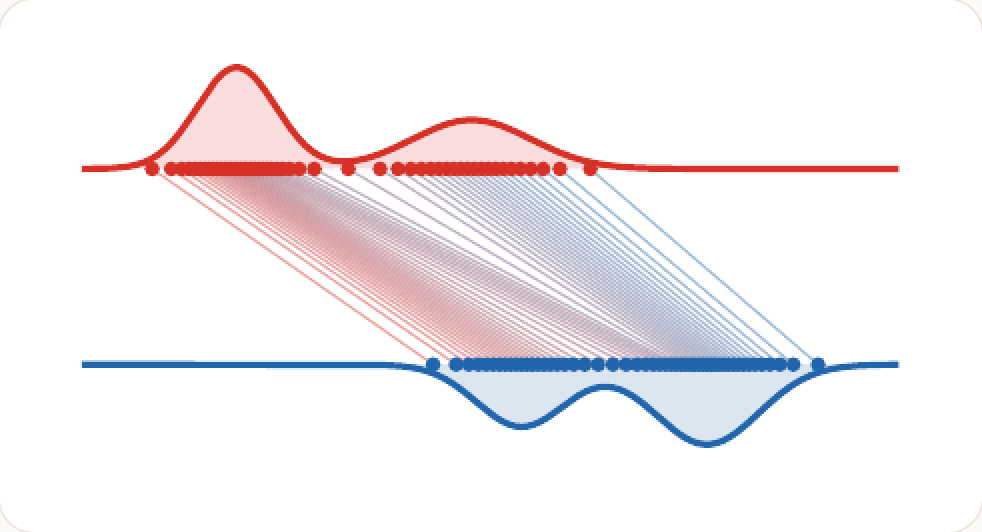
The one-dimensional miracle

For convex costs on the real line, sorting solves the problem.

Monotone matching.
If $x_1 \leq \cdots \leq x_n$ and $y_1 \leq \cdots \leq y_n$, then for every convex increasing cost $h(|x - y|)$,

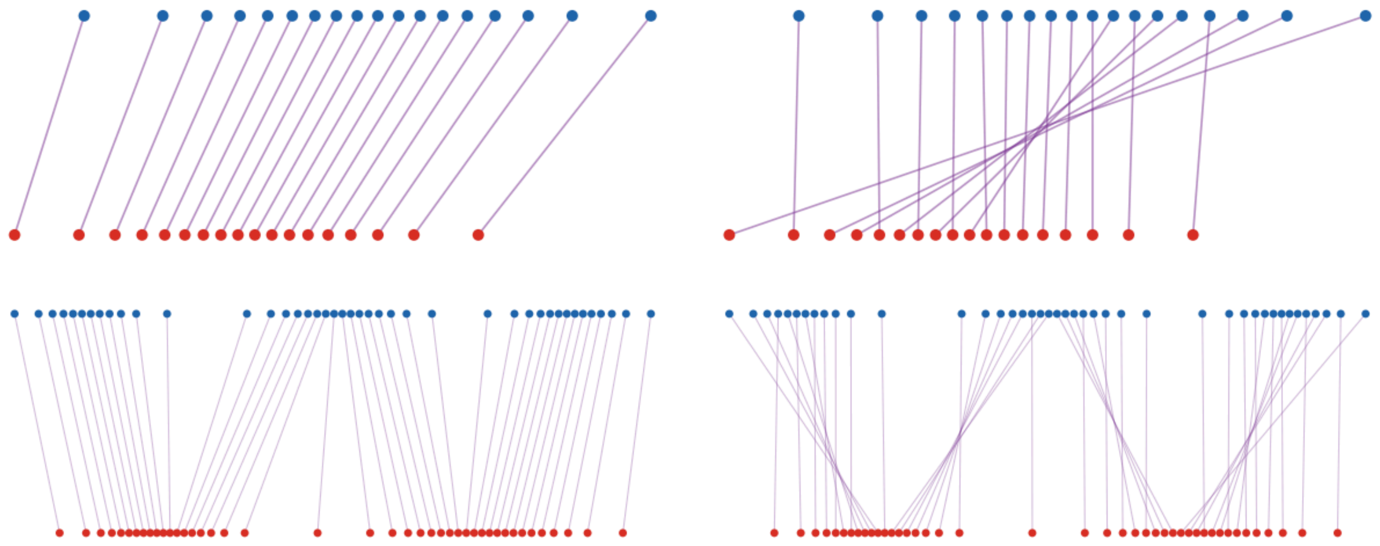
$$\sigma(i) = i$$

is an optimal assignment.



The proof follows from the crossing inequality: replacing two crossing segments by non-crossing ones decreases the cost.

Convex versus concave costs in 1D



Same ordered one-dimensional supports: convex $p = 2$ matchings are monotone, while concave $p = 1/2$ matchings create long crossing exchanges.

For $c_p(x, y) = |x - y|^p$:

- $p > 1$: convexity gives the monotone rearrangement;
- $p = 1$: non-uniqueness can appear;
- $0 < p < 1$: concavity favors different alternating-chain patterns.

The line is special because order is global. Convex costs are solved by sorting; concave costs still exploit the one-dimensional order, but through recursive local matching indicators rather than monotone pairing.

Live panel: one-dimensional costs

points
42

42

source
two

target
three

convex p
2.0

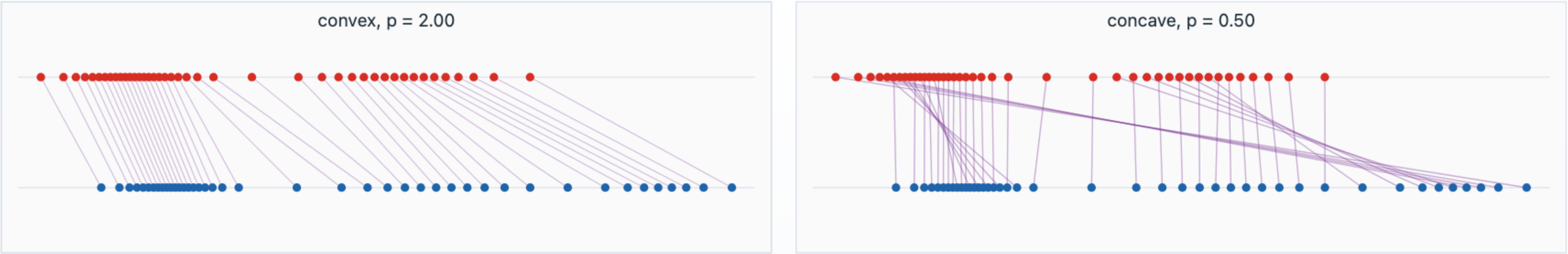
2,0

concave p
0.50

0,50

Reset

Open larger



42 points; convex costs are monotone, concave costs favor long exchanges

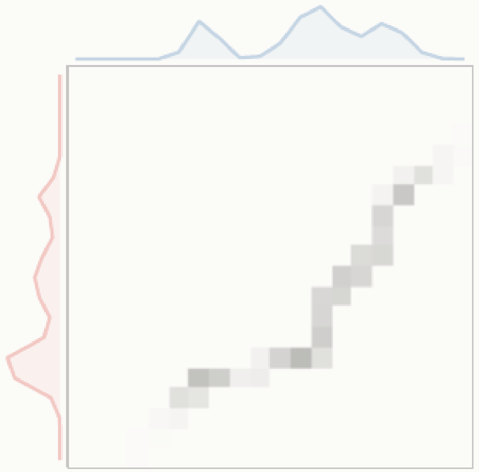
Use the controls to compare how the exponent in $|x - y|^p$ changes the matching geometry.

Roadmap: continuous Monge formulation

1. Monge matching

2. Continuous Monge

3. Kantorovich relaxation



4. Metric structure

5. Applications

Probability measures

Probability measure.
A probability measure α assigns masses $\alpha(A) \in [0, 1]$ to measurable sets $A \subset X$, with $\alpha(X) = 1$.

Discrete

$$\alpha = \sum_i a_i \delta_{x_i}$$

Density

$$d\alpha(x) = \rho_\alpha(x)\, dx$$

Random variable

$$X : \Omega \to X, \qquad \alpha(A) = \mathbb{P}$$

Push-forward

Push-forward.
For a measurable map T from a source space X to a target space Y, the push-forward $T_{\#}\alpha$ is defined by

$$(T_{\#}\alpha)(B) = \alpha(T^{-1}(B)).$$

Equivalently, for all test functions g,

$$\int_Y g(y)\, d(T_{\#}\alpha)(y) = \int_X g(T(x))\, d\alpha(x).$$

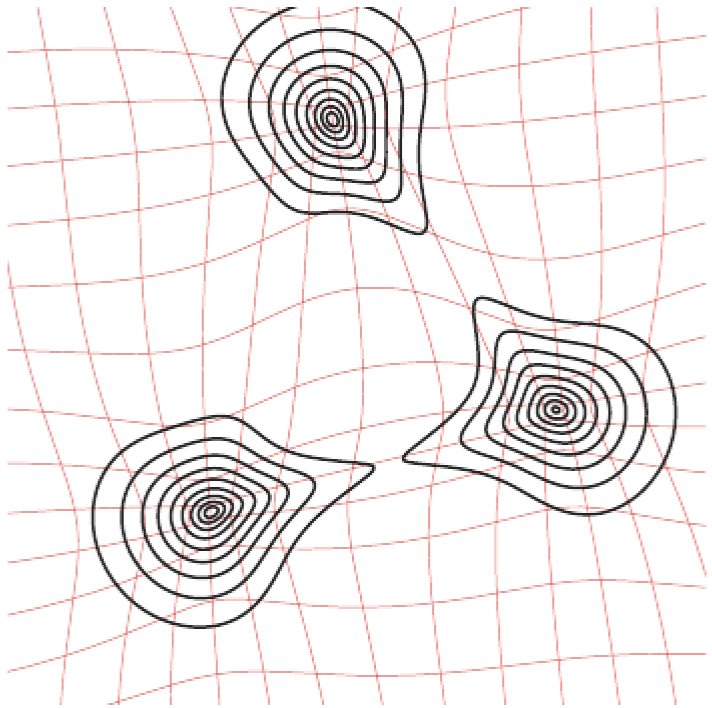
If $\alpha = \sum_i a_i \delta_{x_i}$, then $T_{\#}\alpha = \sum_i a_i \delta_{T(x_i)}$.

Density change of variables

When T is a diffeomorphism between open subsets of $\mathbb{R}^d$,

$$\rho_{T_\# \alpha}(T(x)) = \frac{\rho_\alpha(x)}{|\det DT(x)|}.$$

The Jacobian determinant is not cosmetic: compression creates larger density.



Continuous Monge problem

Monge problem.

Given probability measures α **on** X and β **on** Y, Monge's formulation is

$$\inf_{T_\# \alpha = \beta} \int_X c(x, T(x))\, d\alpha(x).$$

This is a nonlinear constraint: the unknown is a map, and the constraint imposes the entire target distribution.

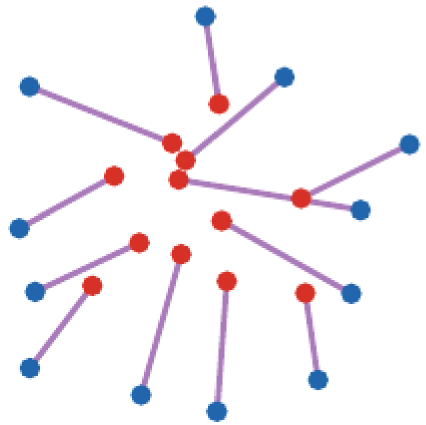
Discrete versus continuous Monge

If

$$\alpha = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i}, \qquad \beta = \frac{1}{n}\sum_{j=1}^{n} \delta_{y_j},$$

then $T_{\#}\alpha = \beta$ means that $T(x_i) = y_{\sigma(i)}$ for some permutation σ.

Thus finite equal-weight matching is a special case of Monge.



Brenier theorem

Brenier theorem.
Let $X = Y = \mathbb{R}^d$ and $c(x, y) = \|x - y\|^2$. If α is absolutely continuous, then the optimal Monge map is unique and

$$T = \nabla\varphi$$

for a convex function φ.

The theorem replaces arbitrary maps by gradients of convex potentials: geometry enters through convexity.

Monge-Ampere equation

If $T = \nabla\varphi$ is smooth and α,β have densities, then

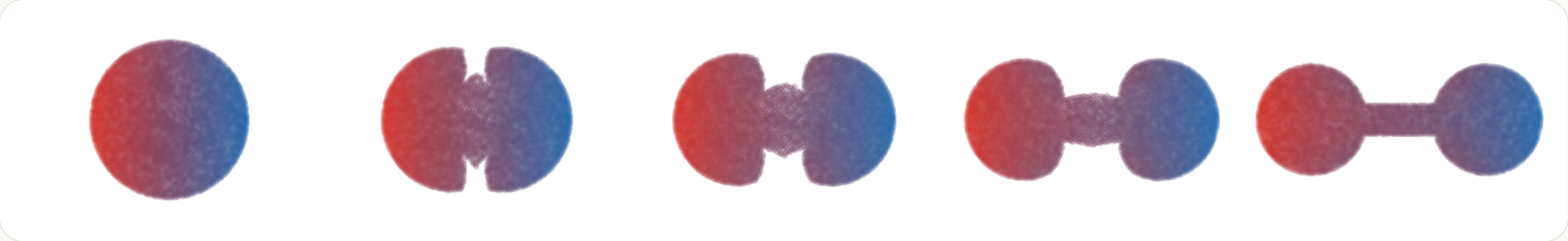
$$\rho_\alpha(x) = \rho_\beta(\nabla\varphi(x))\,\det(D^2\varphi(x)).$$

This nonlinear PDE is the analytic form of the mass conservation constraint.

Regularity is geometric

Brenier maps are well behaved under geometric assumptions, but singularities appear when the target geometry is poor.

Caffarelli principle.
For quadratic transport, convexity and smooth positive densities are central hypotheses behind regularity of the map $T = \nabla\varphi$.



Empirical quadratic OT interpolation from a disk to a connected non-convex two-disk target.

One-dimensional quantiles

CDF and quantile.
For a probability α on $\mathbb{R}$,

$$F_\alpha(x) = \alpha((-\infty, x]), \qquad F_\alpha^{-1}(u) = \inf\{x : F_\alpha(x) \geq u\}.$$

If α has no atoms, the monotone Monge map from α to β is

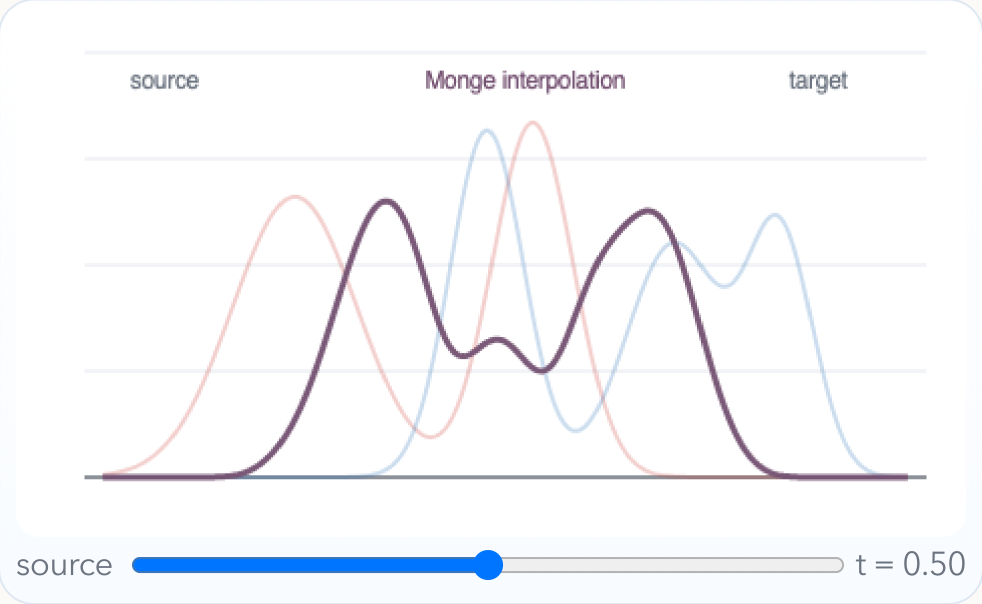
$$T_{\alpha \to \beta} = F_\beta^{-1} \circ F_\alpha.$$

Interactive: quantile interpolation

The 1D Wasserstein geodesic is linear in quantile functions:

$$F_{\alpha_t}^{-1}(u) = (1-t)F_{\alpha}^{-1}(u) + tF_{\beta}^{-1}(u).$$

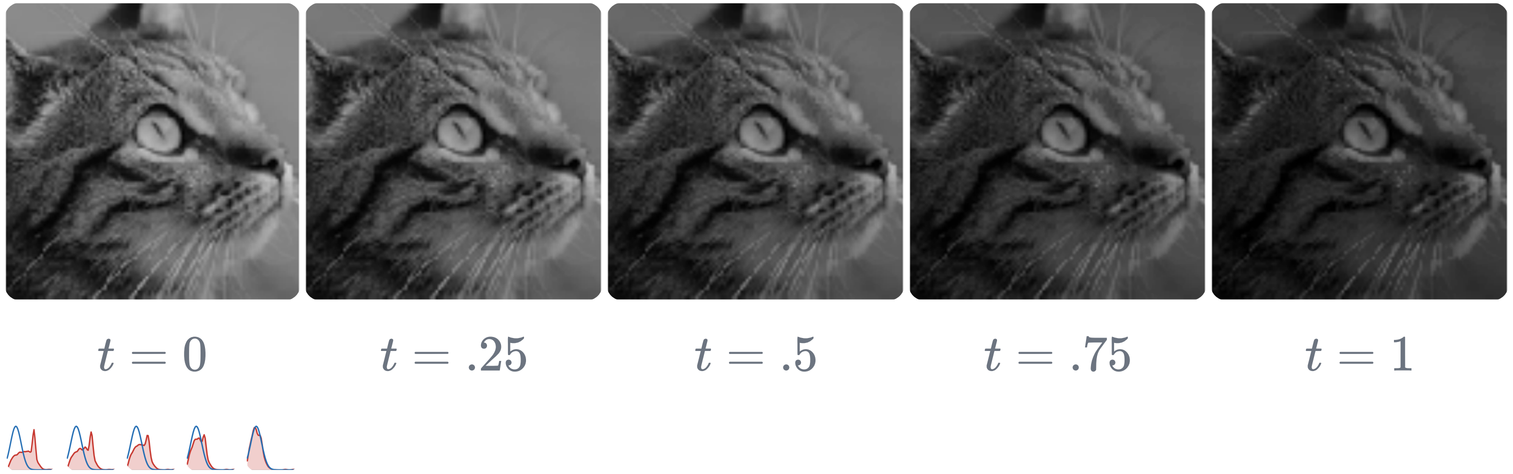
Move the slider to see the density induced by interpolating quantiles.



Live panel: quantile rearrangement

This is the exact one-dimensional Monge solution $T = F_\beta^{-1} \circ F_\alpha$.

Histogram equalization



$t = 0$ $t = .25$ $t = .5$ $t = .75$ $t = 1$

Histogram equalization is a direct application of monotone rearrangement:

$$T = F_\beta^{-1} \circ F_\alpha.$$

Here α is the grayscale distribution of the input image and β is the target grayscale distribution.

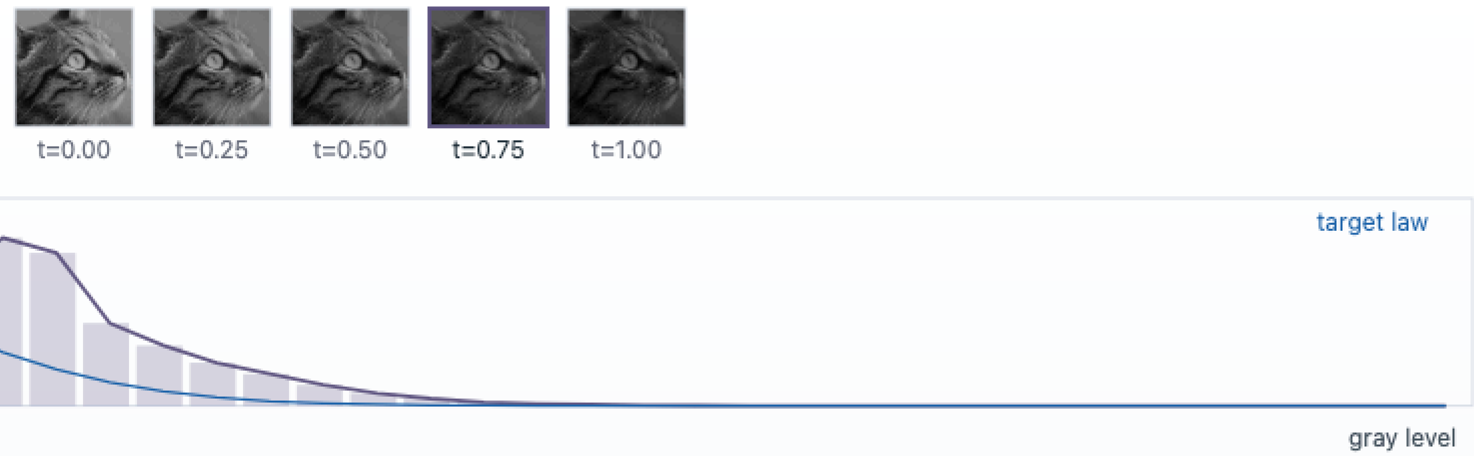
Live panel: histogram equalization

mean
0.18
0,18

sigma
0.105
0,105

t
0.67
0,67

Reset   Open larger



interpolated equalization images; target mean 0.18, sigma 0.105; cat photograph, 180x180

McCann interpolation

Displacement interpolation.

If $T_\# \alpha = \beta$, the Monge interpolation is

$$\alpha_t = ((1-t)\,\mathrm{Id} + tT)_\# \alpha, \qquad t \in [0,1].$$

When T is the quadratic optimal map, this is a constant-speed W_2 geodesic:

$$W_2(\alpha_s, \alpha_t) = |s-t|\, W_2(\alpha, \beta).$$

Live panel: McCann shape interpolation

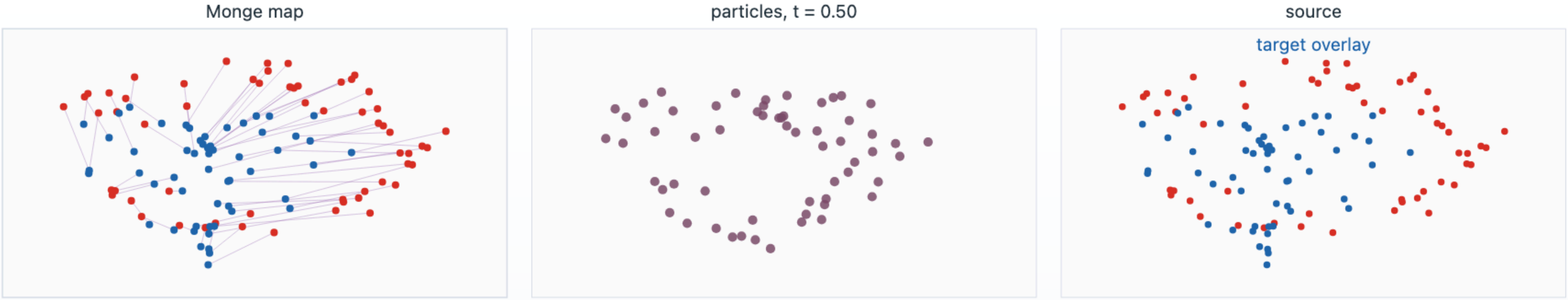
t
0.50
0,50

particles
58
58

source
crescent

seed
2044
2044

Reset | Open larger



58 transported particles; exact assignment used as a small Monge map

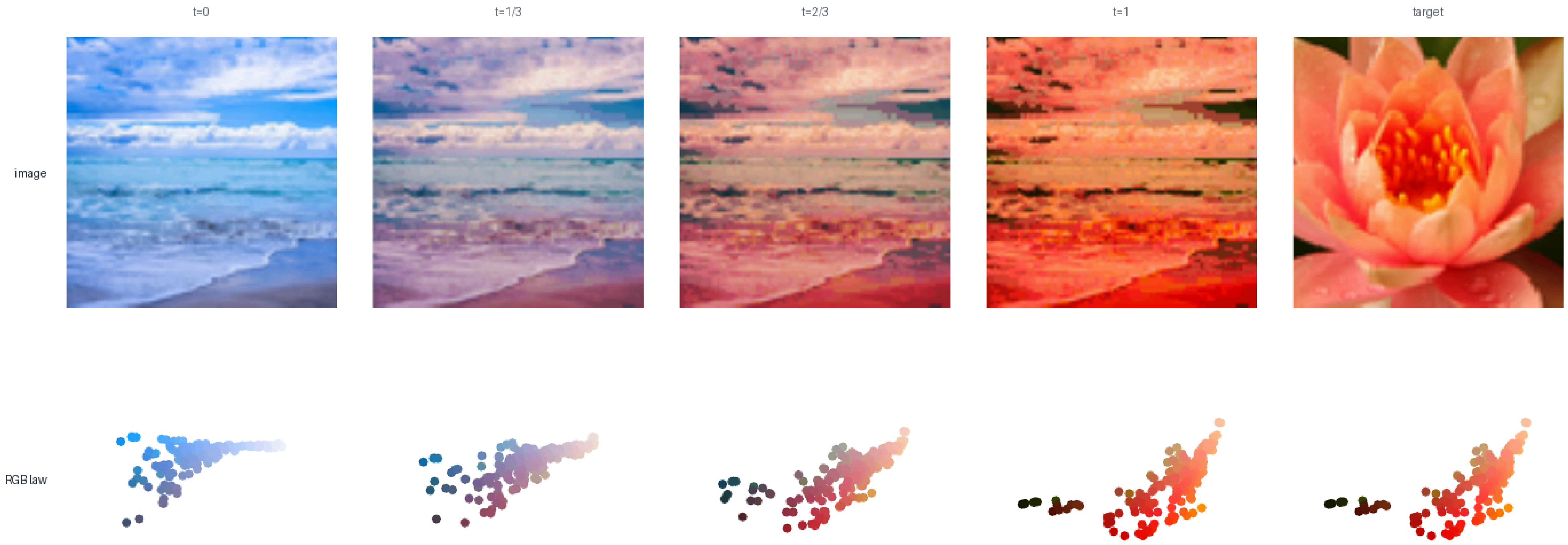
Color transfer as transport

A color image can be viewed as a cloud in RGB space. Transporting one color distribution onto another gives a palette transfer.

For each input pixel color $x \in \mathbb{R}^3$, apply an approximate OT map $T(x)$ in RGB space and display

$$x_t = (1-t)x + tT(x), \qquad \alpha_t = ((1-t)\,\mathrm{Id} + tT)_{\#}\alpha.$$

The top row shows the impact on pixels, while the bottom row shows the moving RGB distribution.



Live panel: color transfer

t
0.62
0,62

resolution
320
320

target
flower

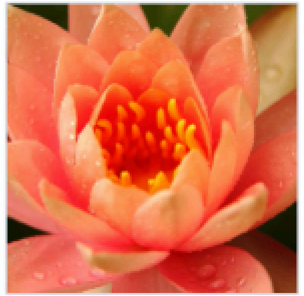
contrast
1.00
1,00

Reset Open larger

source

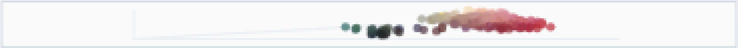
t = 0.62

target



source RGB

transported



target RGB

ranked RGB map with 102400 colors; photograph colors; target flower

Gaussian measures in one dimension

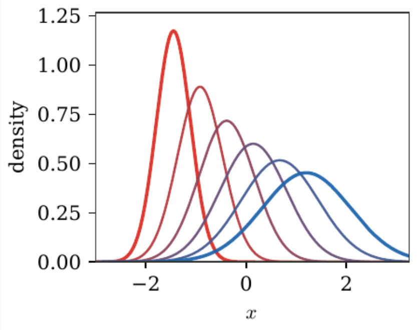
Closed form for 1D Gaussians.

For $\alpha = \mathrm{N}(m_0, \sigma_0^2)$ and $\beta = \mathrm{N}(m_1, \sigma_1^2)$,

$$T(x) = m_1 + \frac{\sigma_1}{\sigma_0}(x - m_0),$$

and

$$W_2^2(\alpha, \beta) = (m_0 - m_1)^2 + (\sigma_0 - \sigma_1)^2.$$



Gaussian measures in dimension d

Gaussian optimal map.
For $\alpha = \mathrm{N}(m_0, \Sigma_0)$ and $\beta = \mathrm{N}(m_1, \Sigma_1)$,

$$T(x) = m_1 + A(x - m_0),$$

where

$$A = \Sigma_0{}^{-1/2}\left(\Sigma_0{}^{1/2}\Sigma_1\Sigma_0{}^{1/2}\right)^{1/2}\Sigma_0{}^{-1/2}.$$

The linear part A is symmetric positive definite: it is a Brenier map.

Gaussian maps and Riccati equations

The Gaussian formula is also a computational statement: the Brenier linear part is the symmetric positive definite solution of a matrix equation.

Riccati viewpoint.
The covariance constraint $T_\# \mathrm{N}(m_0, \Sigma_0) = \mathrm{N}(m_1, \Sigma_1)$ imposes

$$A \Sigma_0 A = \Sigma_1, \qquad A = A^\top \succeq 0.$$

This is an algebraic Riccati equation, whose unique symmetric positive solution is

$$A = {\Sigma_0}^{-1/2} \left({\Sigma_0}^{1/2} \Sigma_1 {\Sigma_0}^{1/2} \right)^{1/2} {\Sigma_0}^{-1/2}.$$

This is one reason Gaussian OT is a useful test case: the nonlinear Monge problem collapses to matrix square roots.

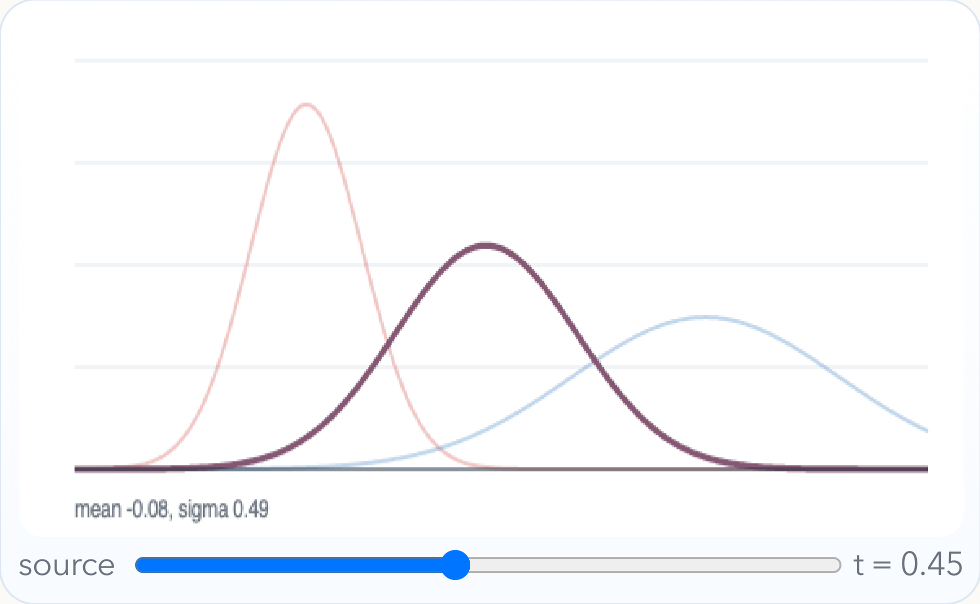
Interactive: Gaussian geodesic

For Gaussian measures, the W_2 geodesic remains Gaussian. In one dimension,

$$m_t = (1-t)m_0 + tm_1, \qquad \sigma_t = (1-t)\sigma_0 + t\sigma_1.$$

In higher dimension, $\Sigma_t = ((1-t)I + tA)\Sigma_0((1-t)I + tA)$.

Move the slider to see the density and the corresponding pointwise affine map.



Live panel: Gaussian maps

t
0.50
0,50

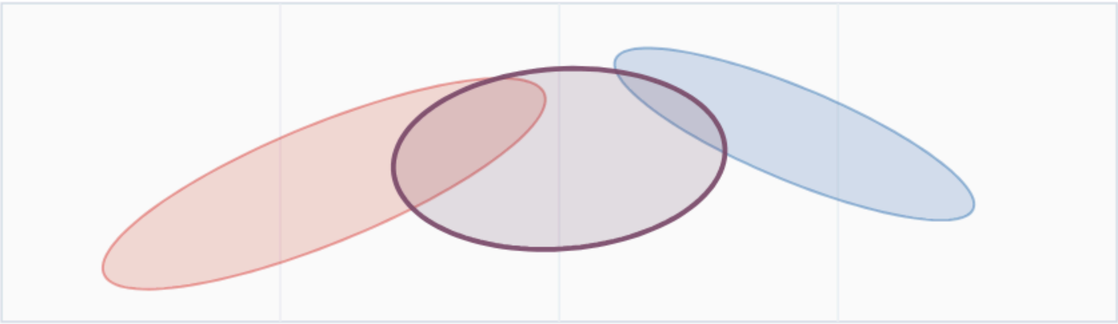
mode
rotated anisotropies

shape
2.5
2,5

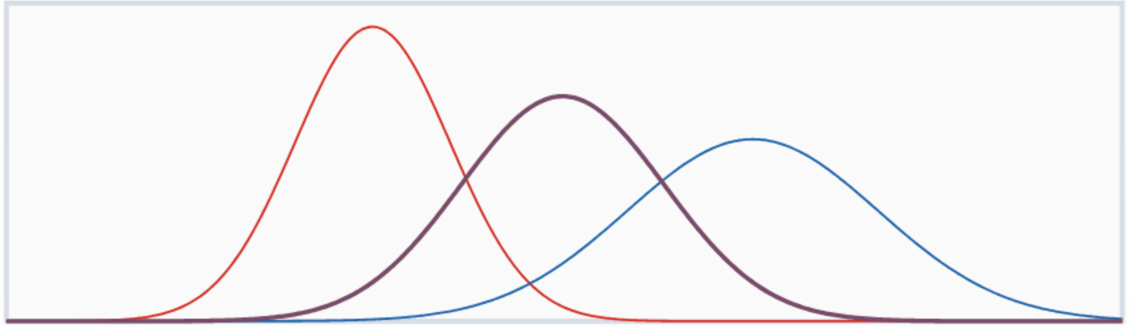
angle
52
52

Reset
Open larger

rotated anisotropies



one-dimensional geodesic



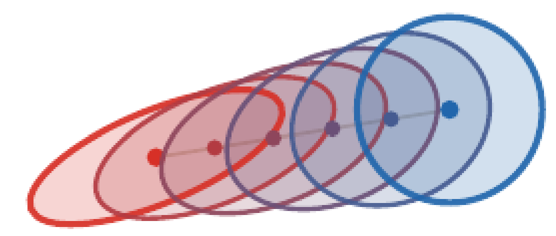
Gaussian geodesic at t = 0.50; covariance follows the Bures interpolation

Bures metric

The covariance contribution to Gaussian W_2 is the Bures metric:

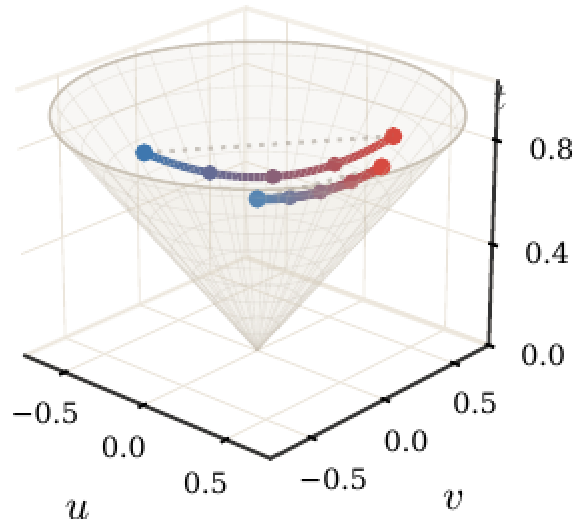
$$W_2^2(\alpha, \beta) = \|m_0 - m_1\|^2 + B^2(\Sigma_0, \Sigma_1),$$

$$B^2(\Sigma_0, \Sigma_1) = \operatorname{tr}\left(\Sigma_0 + \Sigma_1 - 2({\Sigma_0}^{1/2}\Sigma_1{\Sigma_0}^{1/2})^{1/2}\right).$$

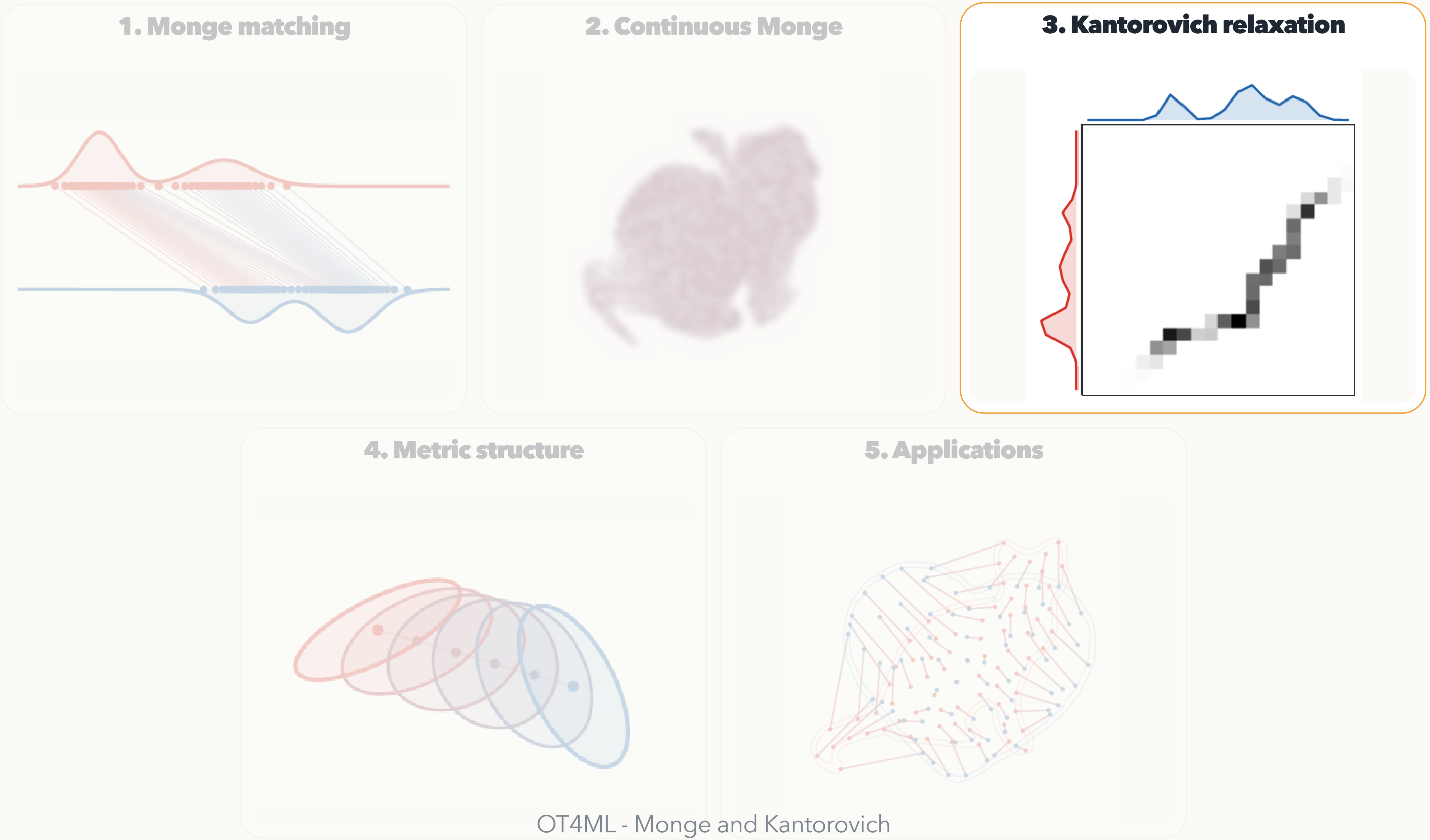


Geometry of covariance matrices

For 2×2 covariances, the positive semidefinite cone can be drawn explicitly. Bures geodesics follow a non-Euclidean matrix geometry.

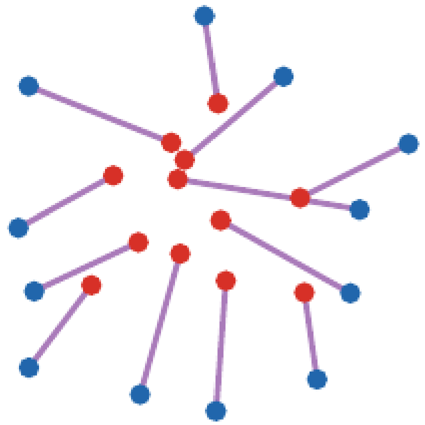


Roadmap: Kantorovich relaxation

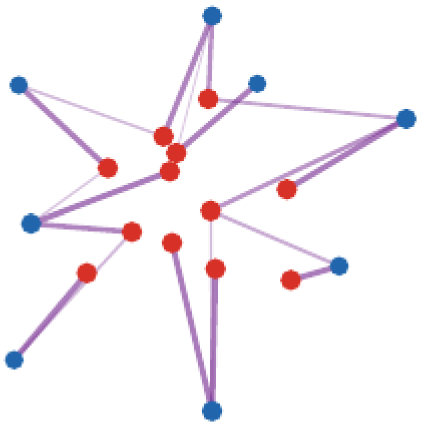


Why relax Monge?

Monge maps can fail to exist when mass must split.



Map-like transport.



Kantorovich allows splitting.

Couplings

Coupling.
A coupling between α and β is a probability measure π on $X \times Y$ such that

$$(P_X)_{\#}\pi = \alpha, \qquad (P_Y)_{\#}\pi = \beta.$$

The set of all couplings is denoted $\Gamma(\alpha, \beta)$.

The product coupling $\alpha \otimes \beta$ always belongs to $\Gamma(\alpha, \beta)$, so the feasible set is nonempty.

Live panel: coupling geometry

points

24

24

plan

splitting

source

disk

target

annulus

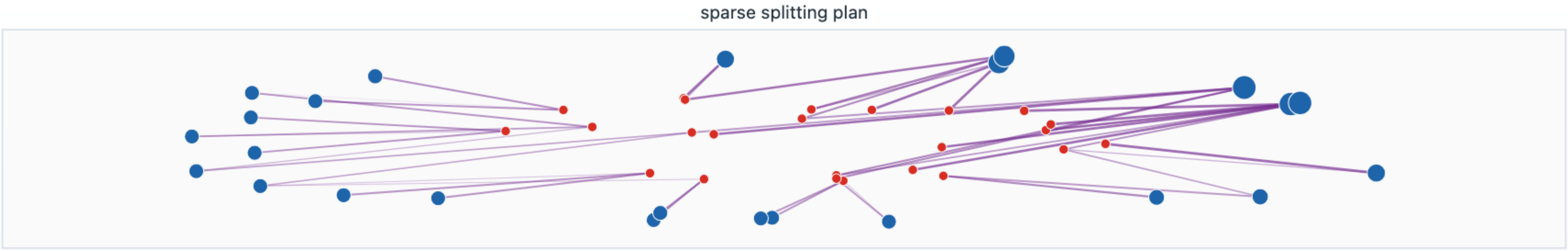
seed

2077

2077

Reset

Open larger



24 sources, 24 targets; 47 positive displayed arcs

Discrete couplings

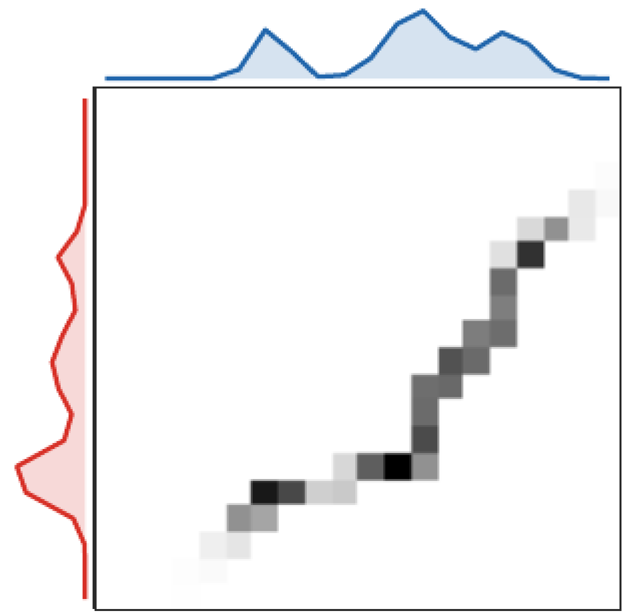
For discrete measures

$$\alpha = \sum_i a_i \delta_{x_i}, \qquad \beta = \sum_j b_j \delta_{y_j},$$

a coupling is a nonnegative matrix P satisfying

$$P\mathbb{1}_m = a, \qquad P^\top \mathbb{1}_n = b.$$

This feasible polytope is denoted $U(a,b) = \{P \geq 0 : P\mathbb{1}_m = a,\ P^\top \mathbb{1}_n = b\}$.



Live panel: transport matrices

bins

48

48

plan

optimal

source

two

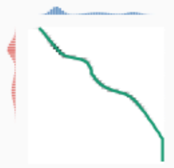
target

three

Reset

Open larger

optimal coupling, 48 bins



93 positive entries

The matrix view makes row and column mass constraints visible.

Three coupling regimes

The same Kantorovich object appears in three numerical regimes.

Discrete, semi-discrete, continuous.
Discrete OT optimizes a matrix P. Semi-discrete OT optimizes Laguerre cells when one measure has a density and the other is finitely supported. Continuous OT studies a measure π on $X \times Y$ and often recovers a **map** through regularity.

The later decks revisit these regimes with duality, Sinkhorn scaling, and gradient-flow interpretations.

Kantorovich problem

Kantorovich formulation.
The relaxed optimal transport problem is

$$\mathcal{L}_c(\alpha, \beta) = \inf_{\pi \in \Gamma(\alpha,\beta)} \int_{X \times Y} c(x, y)\, d\pi(x, y).$$

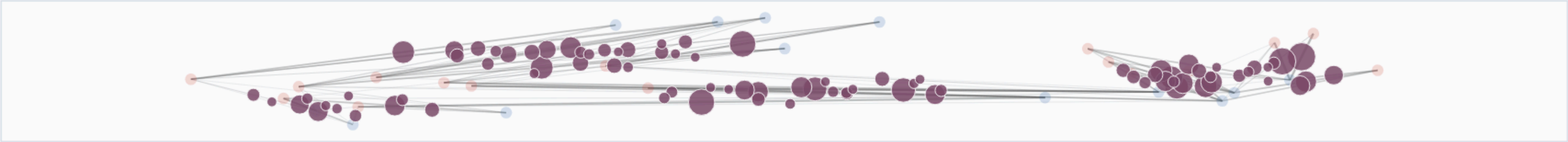
For finite measures, this is the linear program

$$\min_{P \geq 0} \sum_{i,j} P_{i,j} C_{i,j} \quad \text{s.t.} \quad P\mathbb{1}_m = a,\ P^\top \mathbb{1}_n = b.$$

Live panel: transport plans

t	source n	target n	epsilon	seed
0.50	16	13	0.040	2068
0,50	16	13	0,040	2068

Reset Open larger



136 active pairs; t = 0.50, epsilon = 0.040

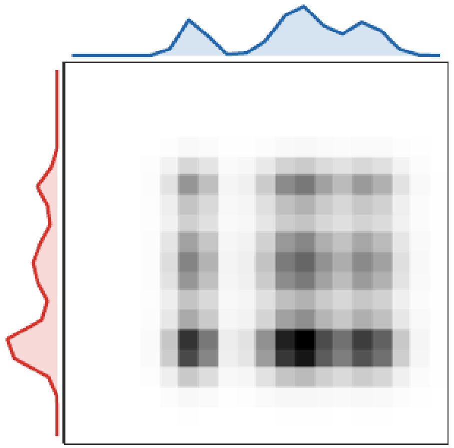
Changing the cost and marginals changes which entries of the plan become active.

Product versus optimal coupling

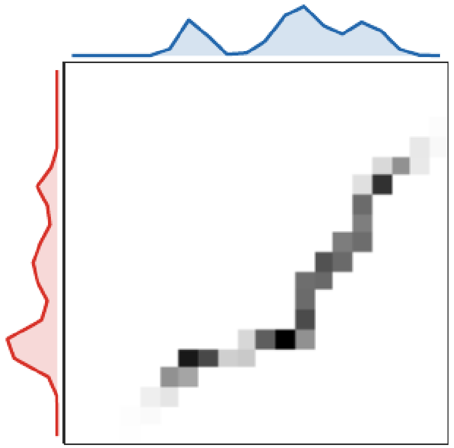
The product plan is the independent coupling:

$$(\alpha \otimes \beta)(A \times B) = \alpha(A)\beta(B), \qquad P^{\otimes}{}_{i,j} = a_i b_j.$$

The OT plan instead minimizes $\langle C, P \rangle$ over $U(a, b)$.



Independent product coupling.



Optimal coupling concentrates near low cost.

Live panel: splitting versus product couplings

sites
14
14

imbalance
1.6
1,6

seed
2059
2059

Reset
Open larger

target imbalance 1.6; sparse plan has 27 positive arcs

The product coupling is feasible but usually diffuse; the optimal plan concentrates on cost-effective pairs.

Exact relaxation for equal weights

Birkhoff-von Neumann theorem.
The set of bistochastic matrices is the convex hull of permutation matrices:

$$\{P \geq 0 : P\mathbb{1} = \mathbb{1}/n, P^\top \mathbb{1} = \mathbb{1}/n\} = \mathrm{conv}\{P_\sigma : \sigma \in \mathfrak{S}_n\}.$$

Here $(P_\sigma)_{i,j} = \frac{1}{n}\mathbf{1}_{\{j=\sigma(i)\}}$ has the prescribed uniform marginals.

Hence, for equal weights, Kantorovich gives the same value as Monge whenever a permutation solution is required.

Extreme points give exactness

The exactness mechanism is purely convex-geometric. For a compact polytope C and a linear functional $\langle A, P\rangle$,

$$\arg\min_{P\in C}\langle A, P\rangle \cap \operatorname{Extr}(C) \neq \varnothing.$$

Indeed, a minimizer exists by compactness. If it is not extreme, write $P = (Q + R)/2$ with $Q \neq R$; linearity shows that Q and R are also minimizers. For a polytope, iterating this splitting reaches an extreme minimizer.

For the bistochastic polytope,

$$\operatorname{Extr}\{P \geq 0 : P\mathbf{1} = \mathbf{1}/n,\ P^\top\mathbf{1} = \mathbf{1}/n\} = \{P_\sigma : \sigma \in \mathfrak{S}_n\}.$$

Thus the relaxed LP has a permutation optimizer.

Live panel: Birkhoff polytope

matrix size

7

7

cycle mass

0.36

0,36

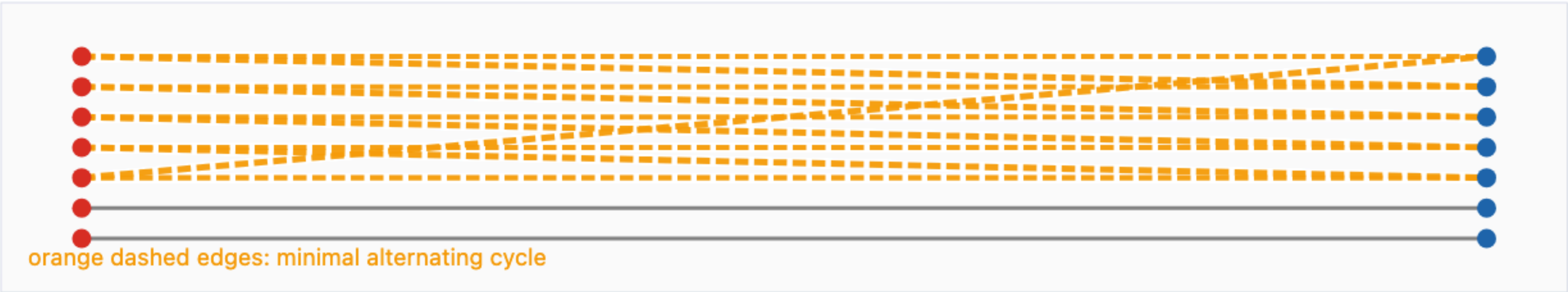
Reset

Open larger

bistochastic plan

0.36	0.64					
	0.36	0.64				
		0.36	0.64			
			0.36	0.64		
0.64				0.36		

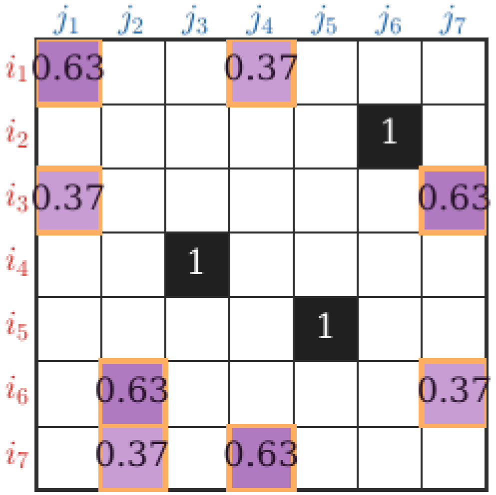
minimal alternating cycle

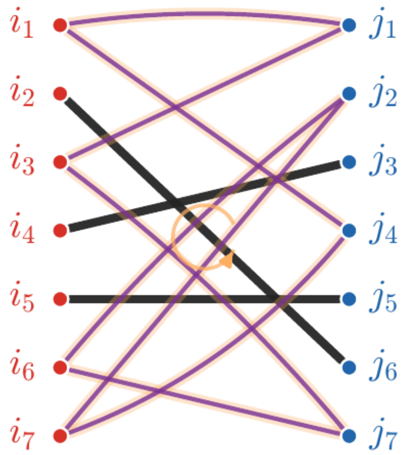


7x7 bistochastic plan; orange cycle has 10 alternating edges with masses 0.36 and 0.64.

Cycle certificate

The proof finds a cycle in the support graph of a non-extreme bistochastic matrix. Moving mass around it decomposes P into two feasible matrices.





Algorithms at a glance

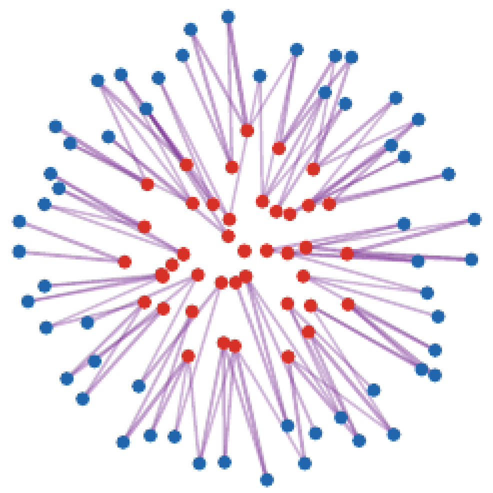
Algorithmic landscape.
- Generic LP or network simplex for discrete OT. - Hungarian or auction methods for assignment. - Sorting for one-dimensional convex costs. - Semi-discrete methods when one measure is continuous and the other is discrete. - Entropic Sinkhorn iterations in the next deck.

The right algorithm depends on the structure of the measures and the cost.

From LP to specialized solvers

Generic linear programming is only one possible computational route; problem-specific algorithms are often decisive.

- assignment: Hungarian and auction methods,
- graph costs: network simplex and sparse flows,
- Euclidean low dimension: geometric structure,
- entropic OT: repeated matrix scaling,
- semi-discrete OT: smooth concave dual maximization.



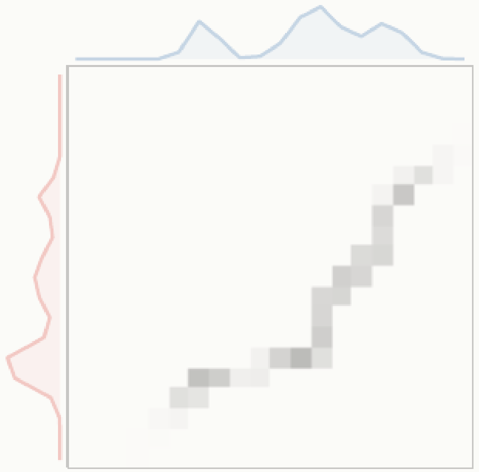
Regularization turns the LP into scalable matrix scaling.

Roadmap: metric structure

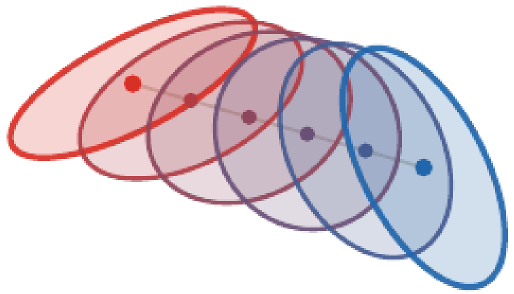
1. Monge matching

2. Continuous Monge

3. Kantorovich relaxation



4. Metric structure



5. Applications

Wasserstein distances

Wasserstein distance.
For $p \geq 1$ and a metric d on X,

$$W_p(\alpha, \beta) = \left(\inf_{\pi \in \Gamma(\alpha,\beta)} \int d(x, y)^p \, d\pi(x, y) \right)^{1/p}.$$

The finite-moment condition is $\int d(x, x_0)^p \, d\alpha(x) < +\infty$ for one, hence every, $x_0 \in X$.

The cost is now tied to the geometry of the ground space.

Metric theorem

Metric property.
On probability measures with finite p-moment, W_p is a distance. In particular,

$$W_p(\alpha, \gamma) \leq W_p(\alpha, \beta) + W_p(\beta, \gamma).$$

Proof idea: glue optimal couplings for (α, β) and (β, γ), then apply Minkowski to the induced pair (X, Z).

Triangle inequality by gluing

The proof mechanism is important: Wasserstein geometry is stable because transport plans can be pasted through their common marginal.

Gluing argument.
Let $\pi^{01} \in \Gamma(\alpha_0, \alpha_1)$ and $\pi^{12} \in \Gamma(\alpha_1, \alpha_2)$. The gluing lemma gives a measure $\eta \in \mathcal{P}(X^3)$ such that $(x_0, x_1)_{\#}\eta = \pi^{01}$ and $(x_1, x_2)_{\#}\eta = \pi^{12}$. Then $\pi^{02} = (x_0, x_2)_{\#}\eta$ is admissible and

$$W_p(\alpha_0, \alpha_2) \leq \left(\int d(x_0, x_2)^p \, d\eta \right)^{1/p} \leq W_p(\alpha_0, \alpha_1) + W_p(\alpha_1, \alpha_2).$$

Discrete gluing formula

The finite proof used in the reference slides is the same idea written as a three-index coupling. If $P \in U(a, b)$ and $Q \in U(b, c)$, set

$$S_{i,j,k} = \begin{cases} \dfrac{P_{i,j} Q_{j,k}}{b_j}, & b_j > 0, \\ 0, & b_j = 0. \end{cases}$$

Then

$$\sum_k S_{i,j,k} = P_{i,j}, \qquad \sum_i S_{i,j,k} = Q_{j,k}, \qquad R_{i,k} := \sum_j S_{i,j,k} \in U(a, c).$$

Summing out the intermediate index j gives the admissible coupling used for the triangle inequality.

Live panel: gluing couplings

outer bins

34

34

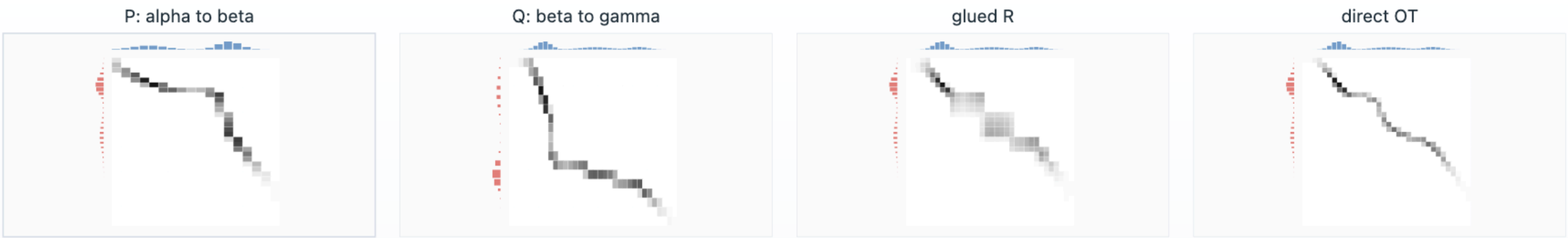
middle bins

18

18

Reset

Open larger



glued matrix R = P diag(1/beta) Q with 18 intermediate bins

Convergence in law

Topology.

On a compact metric space, convergence in W_p is equivalent to weak convergence of probability measures:

$$W_p(\alpha_n, \alpha) \to 0 \quad \iff \quad \int f \, d\alpha_n \to \int f \, d\alpha \quad \forall f \in C(X).$$

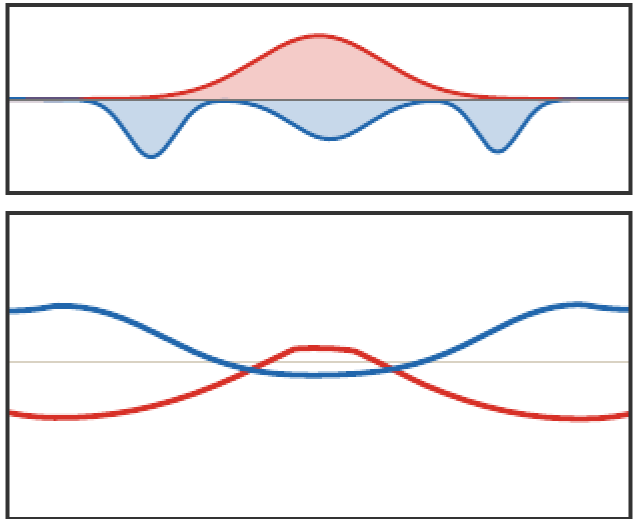
On noncompact spaces, finite moments must also be controlled.

Dual potentials as witnesses

Kantorovich duality exposes potentials as certificates:

$$\mathcal{L}_c(\alpha, \beta) = \sup_{f(x)+g(y)\leq c(x,y)} \int f\, d\alpha + \int g\, d\beta.$$

At optimality, transported mass lives where $f(x) + g(y) = c(x, y)$.

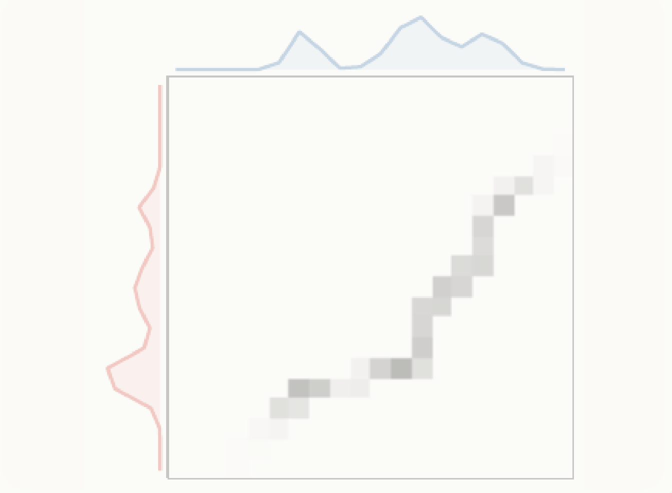


Roadmap: applications

1. Monge matching

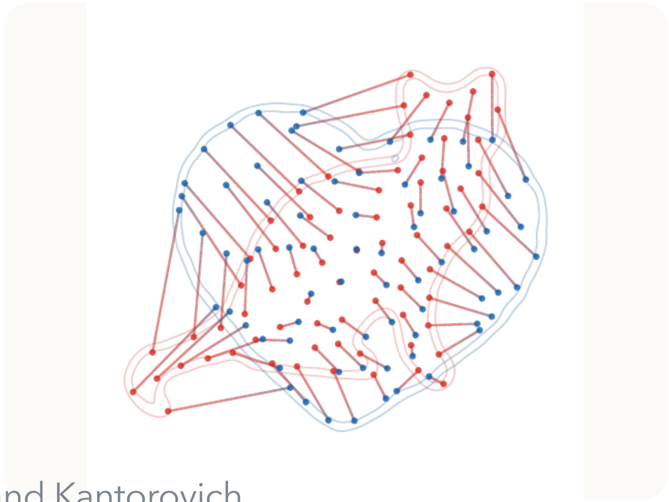
2. Continuous Monge

3. Kantorovich relaxation



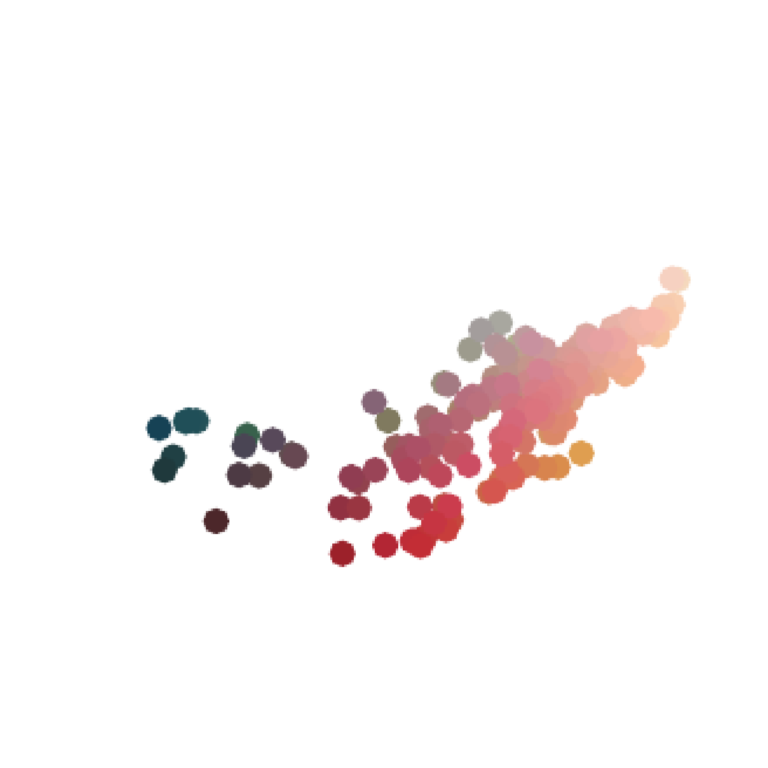
4. Metric structure

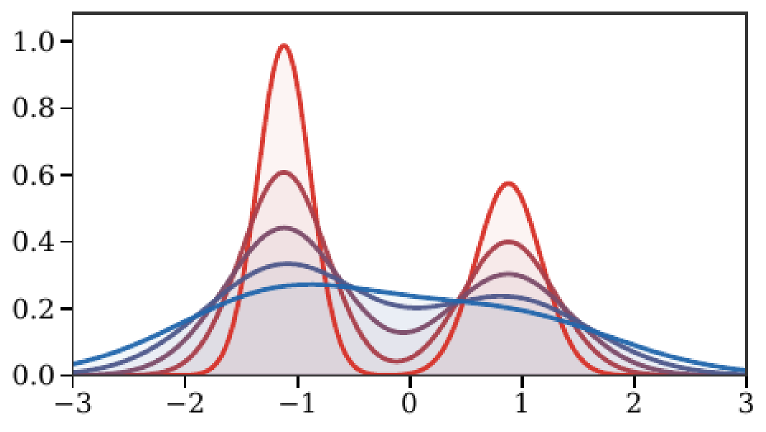
5. Applications



OT application gallery

The same mathematical distance appears in very different data tasks.

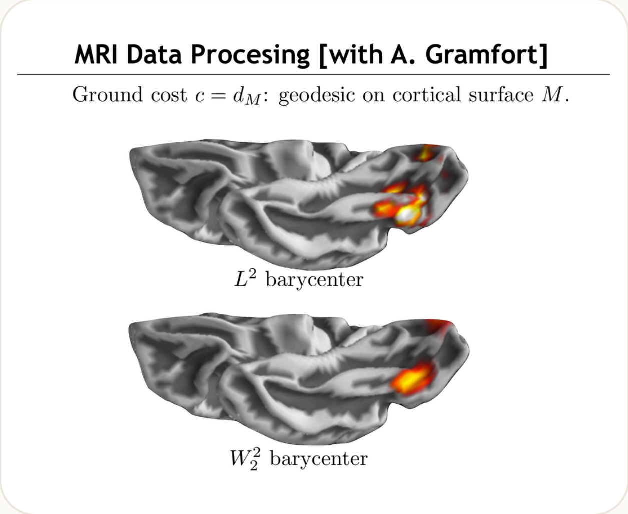


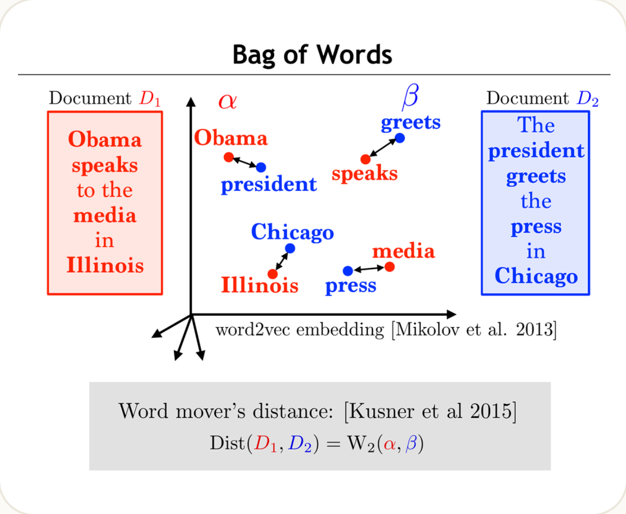


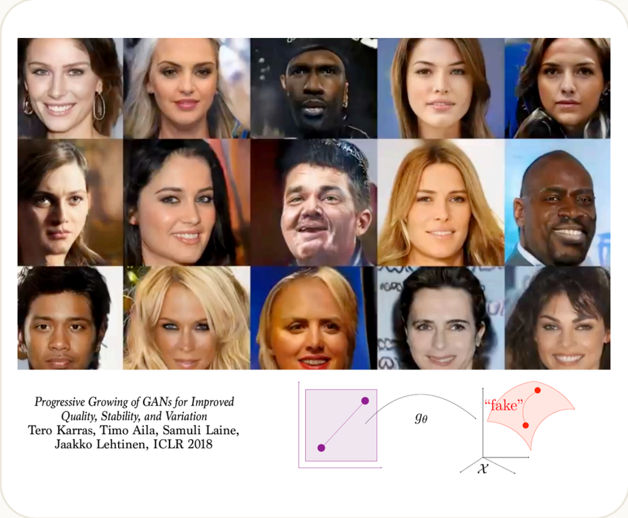
Color transfer, barycenters, and gradient flows all reuse the same core ideas: coupling, geometry, and displacement.

Application examples

Concrete data modalities make the abstract transport formalism easier to recognize.







ATAC-seq and single-cell data, cortical-surface MRI processing, bag-of-words document comparison, and image generation all reduce to comparing empirical laws with geometry.

Quantitative CLT viewpoint

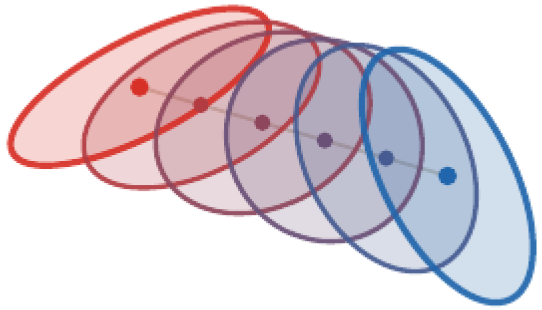
The central limit theorem (CLT) says that rescaled sums converge in law:

$$Y_n = \frac{X_1 + \cdots + X_n}{\sqrt{n}} \Rightarrow \mathrm{N}(0,1)$$

when $\mathbb{E}(X) = 0$ and $\mathbb{E}(X^2) = 1$.

Writing $\mu_n = \mathrm{Law}(Y_n)$ and $\gamma = \mathrm{N}(0,1)$, a Berry-Esseen type estimate in Wasserstein distance is

$$W_1(\mu_n, \gamma) \leq \frac{C\,\mathbb{E}(|X|^3)}{\sqrt{n}}.$$



Wasserstein distances quantify how fast this convergence happens.

Live panel: quantitative CLT

n

16

16

skew

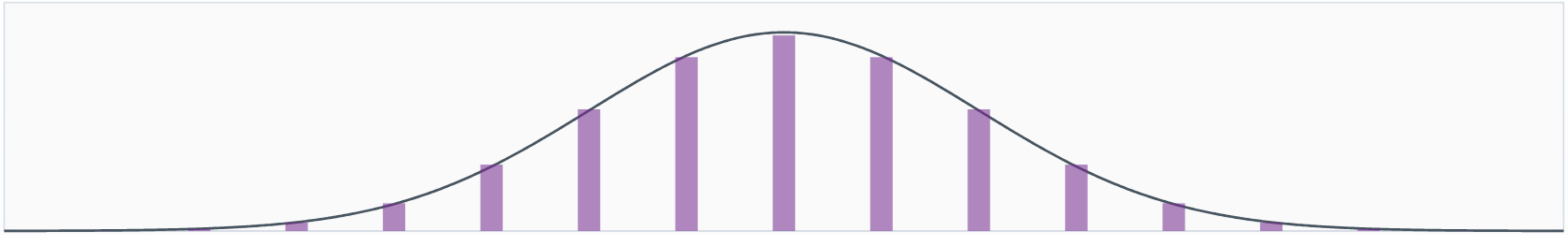
0.00

0,00

Reset

Open larger

normalized Bernoulli sum



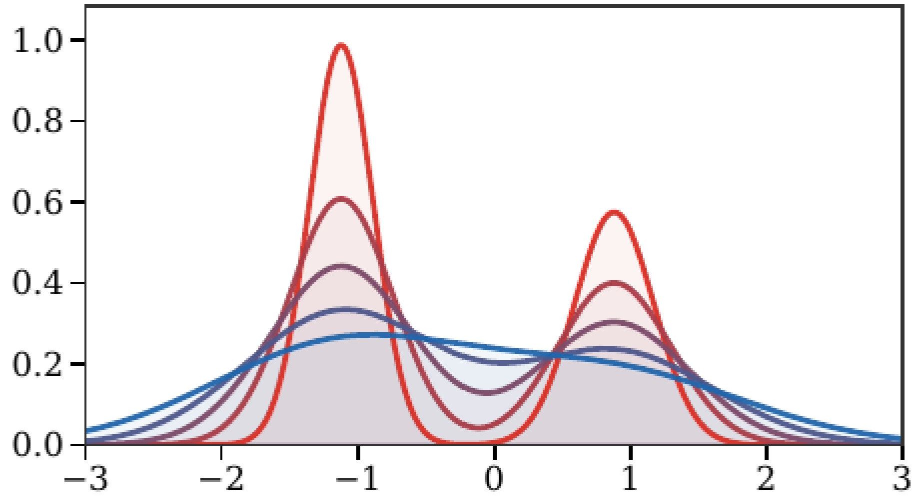
n = 16; skew = 0.00; bar heights are masses divided by lattice spacing 0.500

Gradient flows

Wasserstein geometry turns variational problems over measures into PDEs. The JKO scheme is

$$\alpha^{k+1} \in \arg\min_{\alpha} \frac{1}{2\tau} W_2^2(\alpha, \alpha^k) + \mathcal{F}(\alpha).$$

This is the implicit Euler scheme in the metric space of measures.



Barycenters

Wasserstein barycenter.
Given measures $(\beta_s)_s$ and weights $(\lambda_s)_s$, a barycenter solves

$$\min_{\alpha} \sum_s \lambda_s W_2^2(\alpha, \beta_s).$$

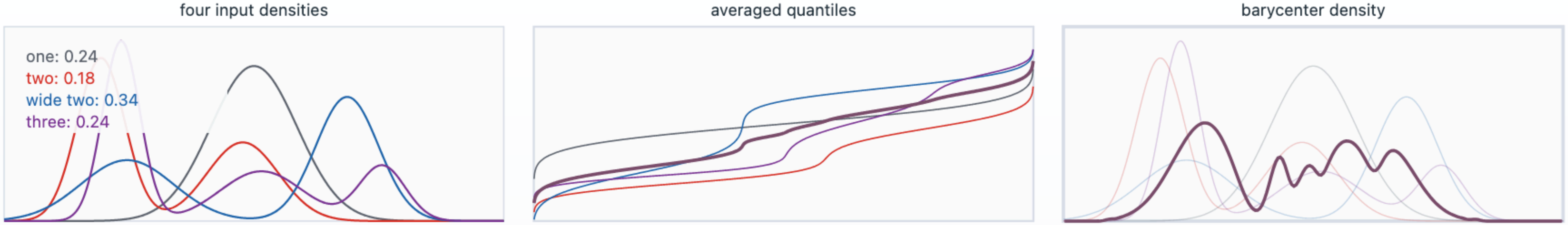
Live panel: Wasserstein barycenters

u weight
0.42
0,42

v weight
0.58
0,58

quantiles
900
900

Reset Open larger



corner weights 0.24, 0.18, 0.34, 0.24; quantile levels 900; barycenter mean -0.349

Barycenters are the first place where transport becomes a geometric averaging operation.

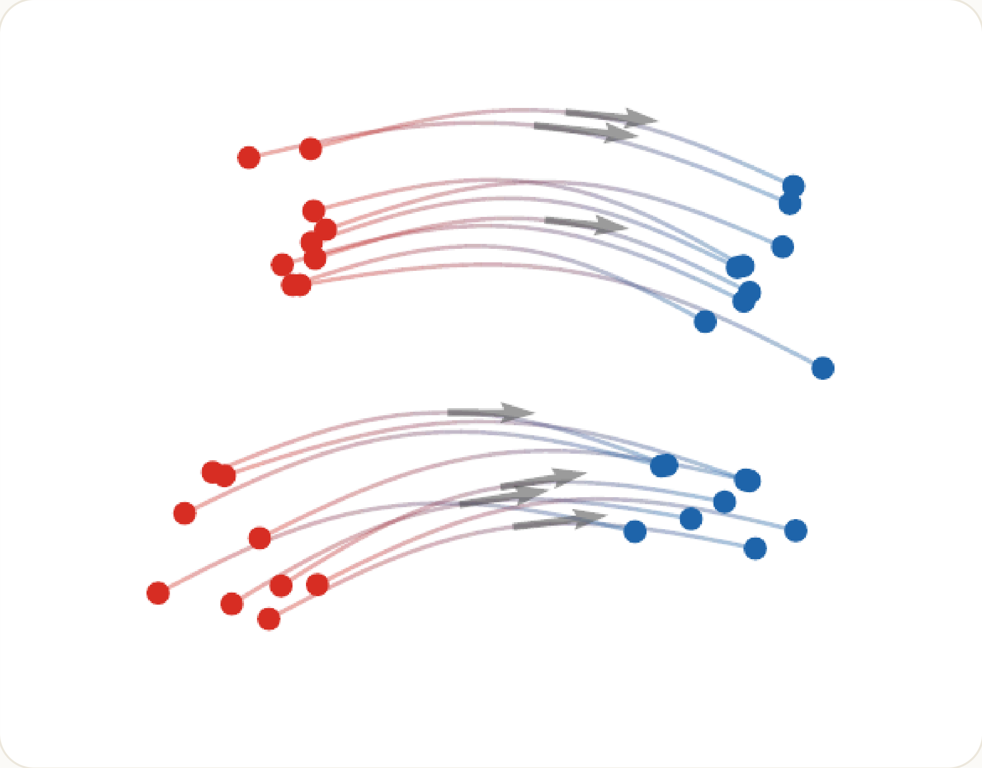
Generative models

One-step generative models seek a map g_θ such that

$$(g_\theta)_\# \zeta \approx \beta,$$

where ζ is a latent distribution and β is the data distribution.

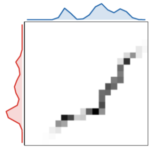
OT, Sinkhorn, MMD, and sliced-Wasserstein losses provide training discrepancies.



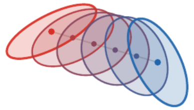
Summary

What this deck establishes.
Optimal transport starts from matching, becomes Monge's map problem, relaxes to couplings, and yields Wasserstein geometry.

Maps
Monge and Brenier



Plans
Kantorovich and LP



Geometry
Wasserstein metrics

Next deck

The next presentation develops entropic regularization:

$$\min_{\pi \in \Gamma(\alpha,\beta)} \int c \, d\pi + \varepsilon \, \mathrm{KL}(\pi \mid \alpha \otimes \beta),$$

leading to Sinkhorn iterations and differentiable transport losses.

Open deck 2: Entropic Regularization →